

Declaring the generator: why `systemictau-declare` exists, and what it is for

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Abstract

The Systemic Tau / RECD programme already has a motor (`systemictau` 4.6), a nested ordinal layer (`nested-recd`), and a machine-checked confrontation between the RECD accumulant T_n and classical thermodynamic time (Module CT in Lean). What it did not have is a **citable operational contract**: a single function that (i) freezes the hyperparameters that scientific runs must share, (ii) reports laboratory time t and gate time T_n as distinct objects, (iii) compares the relational channel ($\Delta\tau_s$, Δexcess^3) with the amplitude channel (variance, lag-1) without declaring a winner, and (iv) **refuses to claim an event when no controls were supplied**. `systemictau-declare` is that contract. Its top-level function is named `declare`, not `predict_outbreak`. This paper explains why that naming is the scientific contribution, what the package may be used for, and what it is forbidden to become.

Keywords: Systemic Tau, RECD, early-warning signals, protocol, false-alarm rate, software contract

1. The gap this package fills

A foundations paper can show that τ_s is an ordinal concordance, that the RECD tick

$$\Delta T_n = g(\tau_s) \delta^{-k} |\tau_s|$$

is not the laboratory second, and that $\text{excess}^3 = 0.6 \text{ Syn} + 0.4 \text{ Surp}$ is a pre-specified Level-3 proxy rather than a Williams–Beer atom. Lean can discharge CT-1–CT-4 / OP-CT-8 with 0 `sorry`. None of that prevents a notebook from retuning θ_3 on the test season, identifying ΔT_n with Δexcess^3 , and painting a red “outbreak risk” in a Streamlit app.

The failure mode is not lack of theory. It is lack of a **frozen entry point**. Epidemiology, Holter analysis, and synthetic maps were accumulating local thresholds. Sensitivity on events without a false-alarm rate was being read as a detector. The public API of `systemictau` still exports `compute_dengue_outbreak_risk` and `compute_market_crash_risk`. Those names teach the wrong verb.

`systemictau-declare` is deliberately small. It does not reimplement Kendall τ , the Feigenbaum gate, or excess^3 . It pins them, hashes the pin, and wraps them in a function whose return value includes a **seal**:

<code>seal.far</code>	Meaning	<code>event_claimed</code>
<code>undefined</code>	no controls	always false
<code>event-only</code>	events without matched controls	always false
<code>estimated</code>	controls present	true only if candidate and surrogate $p < 0.05$

That table is the product. Everything else is instrumentation.

2. Why a package, not another preprint

The corpus already contains theory, Lean, `excess`³ methods, dengue, Holter, RQA, and ontological notes — sometimes twice. Another synthesis would not change what a cardiologist or an epidemiologist can *run*. Three properties had to live in software, not in prose:

1. **Immutability of weights.** 0.6/0.4 is an a-priori design choice. There is no `fit()`. Assigning `Protocol.frozen_v1().weights` raises. Any other pair requires `Protocol.exploratory(...)` and marks the report.
2. **Non-identification in the schema.** ΔT_n and $\Delta excess^3$ are two fields, two docstrings. A JSON dump that silently aliased them would be a scientific bug, not a style issue. Tests assert the alias does not exist.
3. **A FAR that cannot be omitted.** If `controls` is `None`, the seal is **FAR: undefined**. The HTML report prints it in a box. There is no code path from “sensitivity 1.0 on ten events” to a claimed alarm.

A paper can *recommend* these rules. A package can *enforce* them. That is why the artefact is Python.

3. What the package is for (intended use)

3.1 Declaring which generator is in play

Given $X \in \mathbb{R}^{T \times N}$, `declare(X)` returns laboratory t , gate T_n , the sign of the tick $g(\tau_s)$, renormalization depth, and a regime label (`stable` / `chaotic` / `anti-sync` / `intermediate`). In the stable band with frozen depth, Module CT says T_n is locally bi-Lipschitz to a discrete classical clock (CT-1). In recurrent anti-synchronization, no orientation-respecting bridge to a classical clock exists (CT-2, OP-CT-8). The package does not prove those theorems. It **exposes the regime in which they would apply**, so that a methods section can stop saying “time” as if it were unique.

3.2 Comparing two early-warning channels without a trophy

Classical EWS watch variance and lag-1 autocorrelation (critical slowing down). The relational channel watches $\Delta\tau_s$ and $\Delta excess^3$. The scientifically interesting case is already reported in Holter work: the two relational increments can be extreme and sign-concordant with each other *while opposing the amplitude channel*. `declare()` always computes both channels and the two sign agreements. It never returns a “winner”. A later controlled paper can use this API as the frozen detector; it cannot use this API as a scoreboard.

3.3 Writing the paper that can fail

The missing empirical article is not another framework. It is a pre-registered, controlled test of relational versus amplitude channels (Holter with non-event controls; vector surveillance with non-outbreak seasons). That article should call `declare()` and nothing else. If the package is doing its job, a change of θ_3 after seeing the events is impossible without the word **EXPLORATORY** appearing on the report.

3.4 Giving studios a single hook

Existing Streamlit surfaces should call `declare()` or be frozen. They should not keep private copies of the gate. The CLI `recd-declare` is the app: CSV in, one-page HTML out. A wrapper of eighty lines is permitted. A new “ontology studio” is not.

4. What the package is not for

The following uses are in scope for the *programme* and out of scope for *this repository*:

- Closing CT-4C (Schnakenberg bridge). Lean lives in `systemic-tau-formal`. This package cites the pin; it does not import Lean.
- Identifying Σ^{RECD} with Clausius production.
- Identifying excess^3 with Res_{pair} or with a PID synergy atom.
- Claiming retrocausality, or that a receding clock violates the second law. The split clock / production is exactly what allows $\Sigma^{\text{RECD}} \geq 0$ when T_n decreases.
- Replacing the SI second.
- Deploying a dengue dashboard to an agency before a paper with a defined FAR.
- Finance “crash risk” or any red-button predictor. Those names are the anti-pattern this package exists to starve.

A run that needs different B , a permutation null, or a different window is legal. It is not `frozen_v1`. It is `Protocol.exploratory`, and the report says so.

5. Architecture (so the paper can be audited)

<code>systemictau-declare</code>	
— <code>protocols/frozen_v1.yaml</code>	pin + hash payload
— <code>protocol.py</code>	<code>Protocol.frozen_v1 / .exploratory</code>
— <code>run.py</code>	<code>declare()</code>
— <code>seals.py</code>	FAR logic
— <code>report.py</code>	one-page HTML
— <code>cli.py</code>	<code>recd-declare</code>
— <code>tests/</code>	hash, weights, seal, non-ID, G0

Runtime dependencies: `systemictau` (motor: `compute_taus`, `accumulate_time`, RQA) and `nested-recd` ($\Phi_{1,2,3}$, `excess3`). RQA is applied only inside the core-hyper mask ($|\tau_s| < 0.41$ with run length ≥ 7). Inference is on the joint statistic $\sqrt{(\Delta\tau_s)^2 + (\Delta\text{excess}^3)^2}$ against column-wise phase-shuffle surrogates ($B = 99$), not on a difference of means.

The independent-AR fixture G0 is a negative control of the *software*: it must not produce `event_claimed=True`. It is not a proof that real series lack structure.

Feigenbaum δ is the tick scale in `frozen_v1`, not a theorem of this repository. Theorems live in Module CT. CT-4C remains open and is cited as such.

6. How to cite, and what a citation means

Cite the software (this version) if you used `declare()` or `recd-declare` as the detector. Cite Module CT if you claim non-identity of T_n and t as a theorem. Cite the `excess3` methods DOI if you claim the hybrid proxy. Do not cite this package as evidence that a Holter series “proved extramental time”. The package cannot say that. Its most affirmative scientific sentence is:

Under protocol `frozen_v1` (hash ...), the relational contrast ($\Delta\tau_s$, Δexcess^3) had surrogate $p = \dots$ relative to phase-shuffle; FAR was `undefined | event-only | estimated` (= ...). Laboratory lead, if reported, is in t , not identified with ΔT_n .

If that sentence cannot be filled, the run is not yet a result.

7. Conclusion

`systemictau-declare` exists because a second time generator is only scientifically usable when the protocol that estimates it cannot quietly move, and when a missing control cannot be displayed as a detection. It is for declaring t versus T_n , comparing relational and amplitude channels, and sealing FAR. It is not for prediction, ontology, or a fifth ecosystem. The rest of the programme — controlled Holter and dengue papers, CT-4C, studios — should call this API or explain why they will not.

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